

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398004
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Shepherd; Justin

Support apparatus for proppant storage containers

Abstract

A support apparatus includes a frame assembly with an elevated load surface for supporting a modular proppant container in a position above a ground level. A chute assembly is supported by the frame assembly beneath the elevated load surface. A gate actuator has a coupling configured to engage with a gate assembly of the modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly. The support apparatus may include a base frame section having a recessed region beneath the elevated load surface providing a feed station, an on-board subsystem attached to the frame assembly for operating the support apparatus in a stand-alone mode or an in-situ weigh station configured to measure the weight of the modular proppant container supported on an elevated surface in the rectangular container bay.

Inventors:	Shepherd; Justin (Katy, TX)
Applicant:	SANDBOX ENTERPRISES, LLC (Katy, TX)
Family ID:	1000008782566
Assignee:	SANDBOX ENTERPRISES, LLC (Katy, TX)
Appl. No.:	18/136854
Filed:	April 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230249926 A1	Aug. 10, 2023

Related U.S. Application Data

division parent-doc US 17086050 20201030 US 11661291 child-doc US 18136854
us-provisional-application US 62928784 20191031

Publication Classification

Int. Cl.: B65G65/00 (20060101); B65D88/54 (20060101); B65G11/02 (20060101); B65G65/40 (20060101)

U.S. Cl.:

CPC B65G65/005 (20130101); B65D88/54 (20130101); B65G11/026 (20130101); B65G65/40 (20130101); B65G2203/0258 (20130101); B65G2203/042 (20130101)

Field of Classification Search

CPC: B65G (65/005); B65G (11/026); B65G (2203/0258); B65G (65/40); B65G (2203/042); B65D (88/54)

USPC: 177/145

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8668430	12/2013	Oren	414/414	B65D 90/54
9248772	12/2015	Oren	N/A	B65G 65/23
9394102	12/2015	Oren	N/A	B65D 90/58
9694970	12/2016	Oren	N/A	B65D 90/58
11191521	12/2020	Freiburg	N/A	A61B 8/4477
2018/0002120	12/2017	Allegretti et al.	N/A	N/A
2019/0106273	12/2018	Hess et al.	N/A	N/A
2020/0346880	12/2019	Oren	N/A	B65D 90/34

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017/014768	12/2016	WO	N/A

OTHER PUBLICATIONS

Examination report No. 1 issued in couterpart Australian Patent Application No. 2023204137, mailed Sep. 12, 2024 (4 pages). cited by applicant

Primary Examiner: Saint Surin; Jacques M

Attorney, Agent or Firm: Osha Bergman Watanabe & Burton LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Division of application Ser. No. 17/086,050 filed on Oct. 30, 2020, which claims the benefit of U.S. Provisional Application No. 62/928,784, filed on Oct. 31, 2019. The entire disclosures of the above-referenced applications are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates a proppant discharge system for delivering proppant from a bulk storage container, and more particularly relates to a support apparatus configured to locate one or more modular proppant containers in an elevated position, deliver proppant to a feed station and accommodate various functional systems for efficient proppant delivery.

BACKGROUND

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) In the past several years, efforts have been made to improve logistics associated with the transportation, storage and delivery of proppant and other materials used onsite for fracturing operations at an oil/gas well, namely a fracturing site. In any hydraulic fracturing operation, a large amount of such proppant is required. Historically, it was been difficult to effectively transport and store the proppant at the fracturing sites. As a result, efforts have been made to load proppant into a modular container at a sand mine or transload facility, then transport the proppant-filled container to the fracturing site. Once onsite, the proppant-filled containers may be queued up at the fracturing site and proppant dispensed from the containers for use in the fracturing operation. Once emptied, the container may again be queued up at the fracturing site to transported back to the transload facility. Once proppant is material logistics and inventory management systems may be used to provide real-time, accurate information pertaining to the volume/inventory of proppant accessible to a user in a particular region or at a particular location.

(4) Proppant conventionally used in fracturing operations must meet strict specification including moisture and turbidity requirements that require post-mining processes such as washing, screening and drying of the mined frac sand. Once so processed, proppant is relatively “slippery” and can be readily conveyed through handling equipment. Recent efforts to improve fracturing operations have focused on minimizing the post-mining processes of the fac sand by easing the specification for a suitable proppant. Therefore, there is a need to provide improved material handling equipment that is capable of conveying proppant having various characteristics.

SUMMARY

(5) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(6) A support apparatus for unloading a modular proppant container is disclosed herein. In one aspect the support apparatus includes a frame assembly having a base frame section, a plurality of posts extending upwardly from the base frame section and an upper frame section fastened to the plurality of posts in spaced relationship to the base frame assembly. The upper frame section provides an elevated load surface configured to support the modular proppant container in a position above a ground level. The base frame section includes a recessed region beneath the upper frame section providing a feed station. The support apparatus also includes a chute assembly supported by the frame assembly beneath the elevated load surface. The chute assembly includes a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet, and a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating above the recessed region. The support apparatus further includes a gate actuator having a coupling configured to engage with a gate assembly of the modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly.

(7) In another aspect, the support apparatus a frame assembly includes an upper frame section and a plurality of posts fastened to the upper frame section in spaced relationship and extending downwardly therefrom. The upper frame section provides an elevated load surface configured to support the modular proppant container in a position above a ground level. The support apparatus

also includes a chute assembly supported by the frame assembly beneath the elevated load surface. The chute assembly includes a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet, and a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating at a feed station below the elevated surface. The support apparatus further includes a gate actuator including a coupling configured to engage with the gate assembly of a modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly. The support apparatus further includes an on-board subsystem attached to the frame assembly for operating the support apparatus in a stand-alone mode. The on-board subsystem includes a controller configured to switch between a Sleep Mode in which the on-board subsystem is not operational and consumes little to no power and an On Mode in which the on-board subsystem is fully powered and operational.

(8) In another aspect, the support apparatus includes a frame assembly having an upper frame section and a plurality of posts fastened to the upper frame section in spaced relationship and extending downwardly therefrom. The upper frame section includes two longitudinal beams and two cross beams forming a rectangular container bay configured to support the modular proppant container in a position above a ground level. The support apparatus also includes an in-situ weigh station configured to measure the weight of the modular proppant container supported on an elevated surface in the rectangular container bay. The in-situ weigh station includes a scale located at each corner of the rectangular container bay. Each scale has a base plate rigidly attached to the upper frame section and a load cell positioned on top of the base plate. The in-situ weigh station also include a weighing platform having an upper plate resting on top of the load cell at each corner of the rectangular container bay and a rectangular frame extending between the upper plate of adjacent load cells. The in-situ weigh station further includes a load cell processor configured to receive an input data signal from each load cell representing the vertical load between the base plate and the upper plate, compute a total weight on the in-situ weigh station and send an output data signal representing the total weight. The support apparatus further includes a chute assembly supported by the frame assembly beneath the elevated load surface. The chute assembly has a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet and a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating at a feed station below the elevated surface. The support apparatus additional includes a gate actuator having a coupling configured to engage with the gate assembly of a modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly.

(9) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

(2) FIG. 1 is a front-left perspective view showing a support apparatus with a plurality of proppant containers located in an elevated position on the support apparatus;

(3) FIG. 2 is a perspective view of a frame assembly for the support apparatus with FIGS. 2A-2C showing details of joint configuration at various locations (A-C) of the frame shown in FIG. 2;

- (4) FIG. 3 is a plan view of the frame shown in FIG. 2;
- (5) FIG. 4 is a front view of the frame shown in FIG. 2;
- (6) FIG. 5 is a right elevation of the frame shown in FIG. 2;
- (7) FIG. 6 is a detail view of a diagonal frame member;
- (8) FIG. 7 is a front perspective view of the support apparatus according to a first embodiment;
- (9) FIG. 8 is a rear perspective view of the support apparatus shown in FIG. 7;
- (10) FIG. 9 is a rear-right side perspective view of the support apparatus shown in FIG. 7;
- (11) FIG. 10 is a front-left perspective view of the support apparatus shown in FIG. 7
- (12) FIG. 11 is a rear-left perspective view of the support apparatus according to a second embodiment;
- (13) FIG. 12 is a front perspective view of the support apparatus shown in FIG. 11
- (14) FIG. 13 illustrates a main control panel with on-board systems of the support apparatus;
- (15) FIGS. 14-15 illustrate a wireless vision subsystem for monitoring the operation of the support apparatus;
- (16) FIGS. 16-17 illustrate an electrical grounding system for the support apparatus;
- (17) FIGS. 18-21 illustrate a weigh scale system for measuring the weight of a proppant container supported on the support apparatus; and
- (18) FIGS. 22-24 illustrate a remote operator console for monitoring and operating the support apparatus.
- (19) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(20) The following detailed description is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. Furthermore, there is no intention to be bound by any theory presented in the preceding background of the invention or the following detailed description.

(21) With reference to FIG. 1, a plurality of proppant containers **10** are shown supported on a support apparatus **12** in an elevated position above ground level. The proppant containers **10** provide for the bulk storage of proppant or fracturing sand and are readily transportable to and from a well site. A presently preferred embodiment of each proppant container **10** is disclosed in U.S. Pat. No. 9,809,381 to Oren et al., which is expressly incorporated by reference herein. As Oren et al. describe therein, each proppant container includes a gate assembly which is selectively positionable between a closed position and an open position for metering proppant sand from the proppant container.

(22) The support apparatus **12** includes a frame assembly **14** defining an elevated load surface **16** for the proppant containers **10** and a chute assembly **18.1, 18.2, 18.3** (collectively **18**) located beneath for each of the proppant containers for directing proppant discharged from the proppant containers to a feed station. The feed station is a target site such as a blender hopper or proppant conveyor assembly for further handling of the proppant in the fracturing operation. One skilled in the art should recognize that the support apparatus **12** as described herein may be configured for one or more modular proppant containers. For example, in the embodiment illustrated in FIG. 1, the support apparatus **12** is configured with three bays or sections **12.1, 12.2, 12.3** for supporting three proppant containers **10.1, 10.2, 10.3** and therefore include three chute assemblies **18.1, 18.2, 18.3**.

(23) The support apparatus **12** also includes various on-board subsystems **20** associated with the frame assembly **14**. The on-board subsystems **20** may include power supply subsystem having solar panels and/or battery banks as well as power conditioning circuitry for electrically powering the support apparatus, a hydraulic subsystem having a hydraulic pump, a fluid storage tank and a hydraulic for manipulating components of the support apparatus, and a vision subsystem for remotely monitoring the operational state of the support apparatus. These on-board subsystem also

include subsystem controllers for operating the on-board subsystems **20** of the support apparatus **12**. These on-board subsystems **20**, which will be described in greater detail below, enable the support apparatus **12** to be used in a stand-alone or self-sufficient mode of operation that does not require separate or remote utilities such as an external power supply or external pneumatic or hydraulic power source. In this way, the support apparatus **12** physically supports modular containers for discharging proppant to a feed station and also operationally supports the functions of the process by providing electrical, pneumatic and hydraulic power for the support apparatus **12**. (24) With reference now to FIGS. **2-6**, the frame assembly **14** is fabricated using tubular steel members including vertical posts **30**, longitudinal beams **32** and cross beams **34** welded together in a three-dimensional rectangular configuration. Diagonal beams **36** extend between the vertical posts **30** and the upper longitudinal beams **32.1**, **32.2**, **32.3** (collectively referred to as **32**) as well as the vertical posts **30** and the upper cross beams **34.1**, **34.2**, **34.3** (collectively referred to as **34**) to enhance the rigidity of the frame assembly **14** by triangulating the joints between these tubular steel members.

(25) An upper frame section includes upper longitudinal beams **32** and the upper cross beams **34** providing the elevated load surface **16** for supporting the proppant containers **10**. The lower longitudinal beams **32** and lower cross beams **34** define a base frame section or simply base **38** of the frame assembly **14**. The right and left bays **12.1**, **12.3** include a pair of intermediate longitudinal beams **401** extending between the lower cross beams **34** to provide additional floor support at the base of the support apparatus **12**. As illustrated in the figures, the middle bay **12.2** includes a pair of intermediate cross beam **40** secured to a top surface of the lower longitudinal beams **32.2** and extending therebetween. The intermediate cross beams **40c** are configured to receive a pair of forks extending from a fork lift vehicle for lifting and locating the support apparatus **12** at a well site. In another embodiment, the intermediate cross beams **40c** may be secured to a bottom surface of the upper longitudinal beams **32.2** in the middle bay **12.2**. A sheet material **42** may be fastened to the base **38** on the top of the beams **32**, **34**, **401**, **40c** forming a floor of the support apparatus **12**. As best seen in FIG. **3**, the longitudinal beam **32.2** of the middle bay **12.2** is located inboard relative to the longitudinal beams **32.1**, **32.3** of the right and left bays **12.1**, **12.3** to provide a cut-out or recessed region **44** in the frame assembly **14**. This recessed region **44** is configured to receive a hopper for fracturing equipment such as a blender or similar equipment (not shown). In this regard, the recessed region provides a feed station for proppant being discharged from the proppant containers **10**. In other words, the frame assembly **14** has a recessed region **44** formed therein so that the support structure can be located at least partially above the feed station.

(26) The frame assembly **14** may optionally include additional structural elements such as tabs **46** shown in detail **2A** and **2B** for attached items like sheet material **42** to the beams **32**, **34** or cross members **48** to provide additional rigidity and/or attachment locations for the on-board subsystems **20** of the support apparatus **12**. As illustrated in the figures, the sheet material **42** is a metal flooring grid, however one skilled in the art will appreciate that the sheet material **42** may be formed with other materials such as a plastic or composite material and/or be configured in a manner other than a flooring grid such as a diamond plate or other anti-slip sheet material. As presently preferred, the frame assembly **14** further includes a pair of outriggers **50** extending from one of the lower longitudinal beams **32** beyond the base **36**. For example, as best seen in FIGS. **8-9**, the outriggers **50** extend from the rear lower longitudinal rail of right and left bays **12.1**, **12.3**. A leveling jack **52** is disposed at the end of each outrigger **50** and has a height adjustable foot **54** that can be positioned to stabilize and level the frame assembly in a stationary position. As illustrated, the leveling jacks **52** is hand operated to raising and lowering the foot **54** relative to the outrigger **50**. One skilled in the art should, however, recognize that the leveling jack may be motorized or hydraulically actuated for raising and lowering the foot **54**.

(27) With reference to FIGS. **7-10**, the chute assembly **18** and the on-board subsystems **20** will be further described. As previously explained, there is a chute assembly **18.1**, **18.2**, **18.3** associated

with each bay **12.1**, **12.2**, **12.3** respectively. The various structure and function of each chute assembly is sufficiently similar that each chute assembly will be described by reference to chute assembly **18.1** shown in FIG. **8**. The chute assembly **18.1** includes a funnel section **56**, a chute section **58** and a gate actuator **60**. The funnel section **56** is formed by a wall **62** tapering from an inlet or opening **64** at the top of the wall subjacent to the elevated load surface **16** to an outlet **66** below the inlet **64**. As illustrated, the funnel section **56** forms an inverted, truncated pyramid but may be formed by other similar shapes that provide a funneling function from the inlet to the outlet. The chute section **58** extends downwardly from a first end **68** at the outlet **66** of the funnel section **56** to a second end **70** opposite the first end **68** and terminating above the recessed region **44** of the frame assembly **14**. As illustrated, the chute section **58** includes an upper chute **72** secured to the funnel section and a lower chute **74** slidably supported on the upper chute **72**. An actuator **76**, in the form of a mechanical slide, is operably coupled between the upper and lower chutes **72**, **74** so that the overall length of the chute section **58** can be adjusted by sliding the lower chute **74** along the longitudinal axis of the upper chute **72**. One skilled in the art should recognize that the actuator **76** may take other forms such as a motor-driven slide, an electric ram, or a hydraulic or pneumatic or electric actuator. In these embodiments, the actuator **76** may include a limit switch for restricting the relative movement of the upper and lower chutes **72**, **74**.

(28) FIGS. **11-12** illustrate an alternate embodiment of a support apparatus **12'** in which the chute section **58'** includes a cylindrical upper chute **72'** extending from the funnel section **56'** at the first end **68'** and a truncated conical lower chute **74'** extending from the upper chute **72'** and terminating at the second end **70'**. As illustrated, the length of the chute section **58'** is not adjustable, but an angular orientation of the chute section **58'** relative to the funnel section **56'** may be adjustable by providing a bearing or similar rotating interface at the first end **68'** between the funnel section **56'** and the chute section **58'**.

(29) In some applications, proppant stored in the container **10** and dispensed with the support apparatus **12**, **12'** may have a higher moisture or turbidity than conventional proppant that has been dried in post-mining operations. It may be beneficial to specifically configure the support apparatus for these circumstances. For example, portions of the support apparatus **12**, **12'** may be fabricated using a stainless steel material or plastic for providing a slipperier surface than if fabricated using mild steel. In particular, the funnel section **56**, **56'** and/or the chute section **58**, **58'** may be fabricated from stainless steel or plastic. Alternately and/or additionally, the interior surfaces of these sections may be coated with a low friction coating such as a PTFE or similar non-stick coating for reducing the coefficient of friction of the interior surfaces.

(30) Additional features may be used to promote gravity feeding proppant from containers **10** positioned on the support apparatus **12**, **12'** to the feed station. One such feature includes a shaker or vibration mechanism **184** operably coupled between the frame assembly **14**, **14'** and the funnel section **56**, **56'** and/or the chute section **58**, **58'** for gently vibrating these components as proppant discharged from the container **10**. For example, in an embodiment as shown in FIG. **11**, the vibration mechanism **184** may include one or more actuators **186** operably coupled between the frame assembly **14'** and the funnel section **56'** for linearly, orbitally or rotationally vibrating the funnel section **56'** to shake loose any proppant that may have become lodged or stalled therein. While the vibration mechanism **184** is only illustrated in the center bay of frame assembly **14'**, one skilled in the art should appreciate that the other bays of frame assembly **14'** or frame assembly **14** may be similarly equipped with a vibration mechanism. Likewise, in an embodiment illustrated in FIG. **11**, an actuator **188** may be operably coupled between the frame assembly **14'** and the lower chute section **74'** to shake loose any proppant that may have become lodged or stalled therein. While actuator **188** is only illustrated in the center bay of frame assembly **14**, one skilled in the art should appreciate that the other bays of frame assembly **14** or frame assembly **14'** may be similarly equipped with a vibration mechanism. Alternately, the actuator **76**, which in reference to FIGS. **7** and **10** is used to adjust the length of the chute assembly **58**, may also be actuated to vibrate the

chute assembly **58** by moving the lower chute **74** relative to the upper chute **72**. Actuators **186**, **188** may be an electro-mechanical, pneumatic, or hydraulic component. In such applications, the funnel section **56**, **56'** and/or the chute section **58**, **58'** may be resiliently supported from the frame assembly **14**, **14'** to accommodate relative movement therebetween. The amount of shaking movement required may be determined based on the moisture content, turbidity and mesh size of the proppant being discharged from the support apparatus **12**, **12'**.

(31) Another such feature may include an aeration mechanism **190** for injecting a quantity of compressed air or similar fluid stream into proppant being discharged from the container **10**. For example, as shown in FIG. **12**, one or more air injectors **192** coupled to a source of compressed air **194** may be located adjacent the funnel section **56'** and/or in the chute section **58'** for transporting proppant being discharged from the container in a fluid stream. In so doing, proppant exiting the container **10** is fluffed up to reduce the density of the transported material thereby reducing the likelihood of clumping or clogging of proppant. Alternately, a feed mechanism **196** may be implemented along the proppant transport path from the container **10** to the feed station. For example, the feed mechanism **196**, represented by the broken line in FIG. **11** may be configured in the funnel section **56'** of the support apparatus **12**, **12'** and/or along the chute section **58'**. In an embodiment, the feed mechanism **196** is an auger device that moves proppant discharged from the container **10** into and through the funnel section **56**, **56'** and/or through the chute section **58**, **58'** to the feed station. In other embodiments, a conveyor belt, paddle wheel, air stream or similar devices may be implemented for transporting proppant from the funnel section **56'** through the chute section **58'**. While aeration mechanism **190** and the feed mechanism **196** are only illustrated in the center bay of frame assembly **14'**, one skilled in the art should appreciate that the other bays of frame assembly **14'** or frame assembly **14** may be similarly equipped with an aeration mechanism or feed mechanism.

(32) With continued reference to FIGS. **10-11** and **15**, the gate actuator **60** is illustrated. The gate actuator **60** includes a coupling **78** supported on a slide or rail **80** positioned at or adjacent to the elevated load surface **16** extending over the funnel section **56**, **56'**. A linear actuator **82** is operably coupled between the frame assembly **14** and the coupling **78**. As best seen in FIGS. **11** and **15**, the coupling **78** is configured as a receptacle having a slot **84** formed in the top end thereof. The slot **84** receives a pin (not shown) on the gate assembly of a proppant container **10** supported on the elevated load surface. With the gate assembly coupled to the coupling **78**, the actuator **82** functions as a drive mechanism for selectively positioning the coupling **78** (and its received pin) to adjust the gate assembly on the proppant container between the closed and opened position and meter proppant therefrom.

(33) As mentioned above, the support apparatus **12** also includes various on-board subsystems **20** attached to the frame assembly **14**. Referring to FIGS. **7-10** and **13**, the support apparatus **12** is provided with power supply system **22** which include two battery banks **86** rigidly secured to the sheet material **42**. The battery banks **86** include one or more batteries and power conditioning (not shown) that is electrically coupled to an electrical service panel **88** on a main control panel **90** supported at an end of the frame **14**. The service panel **88** may be electrically coupled to the electrical components of the support apparatus **12** for providing primary or auxiliary electrical power thereto. With reference now to FIGS. **16-17**, the power supply system **22** also includes an assembly **114** to electrically ground the support apparatus **12**. The assembly **114** includes a grounding reel **116** with a retractable grounding wire **118** electrically connected to the support apparatus **12**. The assembly **114** also includes a portable grounding rod **120** that can be readily pounded into the ground adjacent the support apparatus **12**. A clamp or clip **122** attached to the end of the retractable ground with **118** can be releasably secured to the grounding rod **120** for electrically grounding the support apparatus **12**.

(34) The power supply system **22** may further include one or more solar panels **92** with power conditioning **94** electrically coupled to the battery banks **86** and/or the electrical service panel **88** to

provide electrical power thereto for charging the battery banks **86** and/or for generating primary or auxiliary power. In one embodiment, the solar panels **92** are positionable with respect to the frame **14** between a stowed position and a deployed position. For example, as seen in FIG. **9**, the solar panel **92** is pivotally supported along an upper edge **94** on an axle **96** between structural members of the frame **14** such as vertical posts **30** or diagonal beams **35**. One or more supports **98** extend between the solar panel **92** and a lower portion of the frame **14**. The supports **98** are adjustable for moving the solar panels **92** between a stowed position vertically oriented between the vertical posts **30** and a deployed position angularly extending from the frame **14** as shown in the figures. The supports **98** are adjustable to affect this pivoting movement of the solar panels **98**. For example, the supports **98** may be a linear actuator for extending and retracting the length of the support. Alternately, the support **98** may be rotatably supported on the frame **14** and slidably supported on the solar panel **92**. Latch posts **100** extend upwardly from the base **38** and terminate at a latch **102** that cooperates with the solar panel **92** for securing it in the stowed position. Once the latch **102** is released, the support **98**, which in one embodiment is a pneumatic cylinder, may extend to deploy the solar panel **92**.

(35) One skilled in the art will recognize that the components used to deploy, operate and stow the various components of the support apparatus **12** may be manually adjustable (e.g., height adjustable foot **54**), mechanically adjustable (e.g., chute actuator **76**), electrically adjustable, hydraulically adjustable (e.g., gate actuator **82**) or pneumatically adjustable (e.g., solar panel supports **98**). When implementing an electrically adjustable component, such as an electrical actuator, it is electrically coupled to the electrical service panel **88** via control devices **104.1**, **104.2**, which may be located locally on the main control panel **90** or remotely in a remote operator console **106**. When implementing a hydraulically adjustable component, such as a hydraulic actuator, it is hydraulically coupled to the other components of the hydraulic system including a hydraulic pump **100** in fluid communication with a hydraulic storage tank or sump **112** and a hydraulic system controller **108**.

(36) The on-board subsystems **20** may include a vision subsystem **124** configured to visually monitor the state and operational status of the support apparatus **12**. As shown in FIG. **14**, the vision subsystem **124** includes a camera assembly **126** for monitoring proppant being discharged from the chute assembly **18**. Camera assembly **126** includes a U-shaped support brace **128** extending from the upper chute **72.2** of the middle chute assembly **18.2**. A camera **130** is secured to the support brace **128** and aimed toward the recessed region **44** in the frame so that the second ends **70** of the lower chutes **74** and a hopper (not shown) positioned in the feed station are within a field of view of the camera **130**. As shown in FIG. **15**, the vision subsystem **124** also includes a camera assembly **132** for monitoring the position of the gate actuator **60** in each of the container bays. Camera assembly **132** includes a support beam **134** extending between upper cross beams **34**. A camera **136** is secured to the support beam **134** and aimed toward the gate actuator **60** so that the coupling **76** and the end of the linear actuator **82** are within a field of view of camera **136**. In an embodiment, camera **130**, **136** are weatherproof cameras electrically coupled to the power supply system **22** and configured to capture and wirelessly transmit live video from a low light scene to a remote display **138** (see FIGS. **23-24**). One skilled in the art will recognize that additional camera assemblies may be deployed on and around the support apparatus **12** to visually monitor the state and operational status of the support apparatus **12**.

(37) The on-board subsystems **20** may include an in-situ weigh scale system **140** configured to measure the weight of the containers **10** supported on the support apparatus **12**. As shown in FIGS. **8** and **18-21**, the weigh scale system **140** includes an in-situ weigh station **142.1**, **142.2**, **142.3** for each of the container bays **12.1**, **12.2**, **12.3**. It will be understood that each of the in-situ weigh stations **142** are substantially the same such that only one weigh station needs to be further described herein. The in-situ weigh station **142** includes four scales **144**, one located at each of the corner of the container bay **12**. With particular reference now to FIGS. **18-21**, each scale **144**

includes a lower support or base **146**, a weighing platform **148** and a load cell **150** positioned between the base **146** and the weighing platform **148**. The base **146** includes a base plate **152** rigidly secured between the upper longitudinal beams **32** and upper cross beams **34**, and a gusset plate **154** rigidly secured between the base plate **152** and the vertical post **30**. The load cell **150** is secured to the base plate **152** by bolts or another similar fastening technique.

(38) The weighing platform **148** includes an upper plate **154** resting on top of the load cell **150**. A stacking cone **156** may be welded to the top of the upper plate **154** and configured to engage, locate and stabilize a container **10** loaded into the bay. A rectangular frame **158** includes angle iron members **160** having a horizontal flange that extends above the upper longitudinal beams **32** and the cross beams **34** and a vertical flange set inside the upper longitudinal beams **32** and the cross beams **34** between the upper plate **154** of adjacent load cells **150** within a given bay. In this way, the weighing platform **148** floats on top of the four scales **144** within the in-situ weigh station **142**. A retainer extends from the weighing platform **148** and is configured to prevent the upper plate **154** from lifting off of the load cell **150** when a container is removed from the support apparatus **12**. In an embodiment shown in FIGS. **18-19**, the retainer is a J-shaped catch **162** extending downward from the frame **158** beneath the upper longitudinal and cross beams **32, 34** to impede any significant upward displacement of the weighing platform **148**. In an embodiment shown in FIG. **20**, the retainer is a pin **164** extend downwardly from the bottom of the upper plate **154** and through a hole **166** in the base plate **152**. A stop **168** is formed on an end of the pin **164** opposite the upper plate **154** to impede any significant upward displacement of the weighing platform **148**.

(39) Each of the load cells **150** in a given weigh station **142** generates a data signal based on the vertical load (i.e., weight) between the base **146** and the weighing platform **148**. These data signals are communicated to a load cell processor **170** in a junction box **172** that computes a total weight on a given weigh station **142**, which may be displayed locally at the weigh station **142** or communicated to a remote digital readout **174** (see FIGS. **23-24**).

(40) With reference now to FIGS. **22-24**, the support apparatus **12** may include a remote operator console **106** which is in wired and/or wireless communications with the other subsystems **20** previously described. In the embodiment illustrated in FIGS. **22-24**, the remote operator console **106** includes a wheeled stand or dolly **176** supporting an enclosure **178** having a hood **180**. An umbilical cord **182** extends from the console **106** to the support apparatus **12** and may include electrical cords for providing electrical power to the enclosure **178** and/or data transmission cords for communicating data signals and control signal between the support apparatus **12** and the remote console **106**. As shown in FIGS. **23-24**, the remote console includes a video display **138** for the vision subsystem **124**, the remote digital readout **174** for the weigh scale system **140**, a tablet **104.2** configured to remotely control the hydraulic system controller **108** and an on-board power supply **176**. One skilled in the art will recognize that the remote operator console may be equipped with additional systems used in the support of proppant delivery to a target site at a fracturing operation such as a radio for providing two-way communication, a laptop computer or other local computing device.

(41) When deployed for a fracturing operation near an oil well site, the support apparatus **12** provides an efficient means for establishing the necessary infrastructure to deliver proppant to a hopper in the feed station. In this regard, the support apparatus **12** is located in a predetermined location using a field transport vehicle such as a fork lift truck. Specifically, the support apparatus **12** may be transported to the well site using conventional means such as a rail car or flatbed trailer-truck. The forks of the fork lift truck are positioned into cross beams **38** and the support apparatus **12** is removed and placed at the predetermined location. For example, the support apparatus **12** may be placed next to a blender apparatus such that the blender hopper (not shown) is located in the recessed region **44** of the frame **14**. Next the support apparatus **12** may be stabilized by extending the feet **54** from the leveling jacks **52** at the end of the outriggers **50** to level the support apparatus **12**. The chutes **18** are then positioned so that the second end **72** of the chute section **58** is

above the blender hopper. The solar panels **92** may also be moved from the stowed position to the deployed position and the on-board subsystems **20** powered up and tested for proper functioning. (42) Once so positioned, the field transport vehicle may be used to retrieve a proppant container **10** and locate it on the elevated load surface **18** such that the pin of a gate assembly is received in the coupling **78**. Additional proppant containers **10** may be retrieved and located on the elevated load surface **18** until all the bays of the support apparatus **12** are occupied. The gate actuator **60** is remotely operated to selectively position the gate assembly for each proppant container **10**. Proppant for each container **10** is gravity-fed through the chute assembly **58** for delivery to the blender hopper. When a proppant container **10** has emptied all of the proppant stored therein, the empty proppant container **10** may be removed from the support apparatus **12** and replaced with a filled proppant container **10**.

(43) Because the support apparatus **12** can be operated in a stand-alone mode, the various controllers associated with the on-board subsystems **20** can be configured to reduce power consumption by switching between a “SLEEP” mode in which subsystems are not operational and consume little to no power and an “ON” mode in which the subsystems are fully powered and operational. For example, the hydraulic system and particularly the hydraulic pump **100** consumes a significant amount of power when fully powered and operational. As such, the hydraulic system controller **108** may be configured go into SLEEP mode by shutting down the hydraulic pump **100** when hydraulic pressure is not needed to operate a hydraulic component. Once a control signal is received to operate a hydraulic component, for example when manipulation of the gate actuator **60** is requested, the hydraulic system controller **108** switches to the ON mode and turns on the hydraulic pump **100** to provide hydraulic pressure for operating the hydraulic component. In this way, the hydraulic system may be instantaneously activated to provide on-demand hydraulics, while conserving power when hydraulics are not needed. A similar on-demand activation may be implemented for other subsystems associated with the support apparatus **10**.

(44) Various embodiments and methods have been presented in the foregoing detailed description, it should, however, be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the invention in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment as contemplated herein. It should be understood that various changes may be made in the function and arrangement of elements described in an exemplary embodiment without departing from the scope of the invention as set forth in the appended claims.

Claims

1. A support apparatus for unloading a modular proppant container having a gate assembly comprising: a frame assembly including an upper frame section having a pair of and a plurality of posts fastened to the upper frame section in spaced relationship and extending downwardly therefrom, the upper frame section having two longitudinal beams and two cross beams forming a rectangular container bay configured to support the modular proppant container in a position above a ground level; an in-situ weigh station configured to measure the weight of the modular proppant container supported on an elevated surface in the rectangular container bay, the in-situ weigh station including: a scale located at each corner of the rectangular container bay, each scale including a base plate rigidly attached to the upper frame section and a load cell positioned on top of the base plate; a weighing platform including an upper plate resting on top of the load cell at each corner of the rectangular container bay and a rectangular frame extending between the upper plate of adjacent load cells; and a load cell processor configured to receive an input data signal from each load cell representing the vertical load between the base plate and the upper plate,

compute a total weight on the in-situ weigh station and send an output data signal representing the total weight; a chute assembly supported by the frame assembly beneath the elevated load surface, the chute assembly having a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet, a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating at a feed station below the elevated surface; and a gate actuator including a coupling configured to engage with the gate assembly of a modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly.

2. The support apparatus according to claim 1, wherein the rectangular frame of the weighing platform comprises a horizontal flange extending above the longitudinal beams and the cross beams and a vertical flange set inside the upper longitudinal beams and the cross beams, wherein the weighing platform floats on top of the four scales within the in-situ weigh station.

3. The support apparatus according to claim 1, wherein the in-situ weigh station further comprises a plurality of retainers extending from the weighing platform and configured to engage the upper frame section for preventing the upper plate from lifting off of the load cell when the modular proppant container is removed from the support apparatus.

4. The support apparatus according to claim 3, wherein each retainer comprises a J-shaped catch extending downward from the rectangular frame beneath at least one of the longitudinal beam and the cross beams to impede upward displacement of the weighing platform when the modular proppant container is removed from the support apparatus.

5. The support apparatus according to claim 3, wherein each retainer comprises a pin extending downwardly from a bottom of the upper plate and through a hole formed in the base plate and a stop formed on an end of the pin opposite the upper plate to impede any significant upward displacement of the weighing platform when the modular proppant container is removed from the support apparatus.
